

Practical 2 — Functional space of freshwater fishes

Master FBE · Functional Ecology in R

In this practical you will build a functional space for freshwater fish species using morphological traits from FishMORPH (Brosse et al. 2021, GEB). The data file `fish_traits.csv` contains 7 standardised morphological traits for ~500 species across 5 biogeographic realms.

```
library(FD); library(funspace); library(ggplot2); library(dplyr)

traits <- read.csv("fish_traits.csv", row.names=1)
head(traits)
```

Question 1 — Trait exploration (10 pts)

1a. Check trait completeness (% non-NA per trait). Which trait has the most missing data? Drop species with more than 30% missing traits.

```
# Completeness
colMeans(!is.na(traits))*100

# Filter species
complete_pct <- rowMeans(!is.na(traits))
traits_clean <- traits[complete_pct >= 0.70, ]
```

1b. Plot the pairwise correlation matrix of traits as a heatmap. Identify two pairs of strongly correlated traits ($|r| > 0.7$) and explain the morphological reason.

```
library(reshape2)
cor_mat <- cor(traits_clean, use="pairwise.complete.obs")
cor_melt <- melt(round(cor_mat, 2))
ggplot(cor_melt, aes(Var1, Var2, fill=value, label=value))+
  geom_tile(colour="white")+geom_text(size=2.8)+
  scale_fill_gradient2(low="#c0392b", mid="white", high="#8e44ad", midpoint=0)+
  theme_minimal()+theme(axis.text.x=element_text(angle=45, hjust=1))
```

Question 2 — PCA on quantitative traits (20 pts)

2a. Run a PCA on scaled traits. How many axes are needed to explain 70% of variance?

```
traits_sc <- scale(traits_clean)
pca <- prcomp(traits_sc, scale.=FALSE)
cumsum(summary(pca)$importance[2, ])
```

2b. Plot a biplot (PC1 vs PC2) coloured by realm. Which realm occupies the most unique region? Overlay trait loadings as arrows.

```
library(funspace)
realm_vec <- traits_clean$Realm # if realm column exists
```

```
funspace(x=pca, axes=c(1,2), group.vec=realm_vec,  
        pnt=TRUE, fill.col=c("#2980b9", "#27ae60", "#e67e22", "#8e44ad", "#c0392b"))
```

2c. Interpret PC1 biologically: which traits have the highest absolute loadings? What ecological gradient does PC1 represent?

```
sort(abs(pca$rotation[,1]),decreasing=TRUE)
```

Question 3 — PCoA with Gower distance (15 pts)

3a. Compute Gower distance between species (handles NA automatically). Run a PCoA. Compare the variance explained by PC1/PC2 with the PCA result.

```
d_gower <- gowdis(traits_clean)
pcoa <- cmdscale(d_gower, k=6, eig=TRUE)
eig_pct <- 100*pcoa$eig/sum(abs(pcoa$eig))
cat(sprintf("PCoA - PC1=%.1f%% PC2=%.1f%%\n", eig_pct[1], eig_pct[2]))
```

3b. The Neotropical realm hosts >75% of global fish functional diversity (Toussaint et al. 2016). Test this: compare functional overlap between the Neotropical realm and all others combined.

```
groups <- factor(ifelse(traits_clean$Realm=="Neotropical", "Neotropical", "Other"))
overlap <- fun_overlap(x=pca, axes=c(1,2), group.vec=groups)
print(overlap)
```

3c. Compute functional uniqueness per species. List the 5 most unique species. What morphological traits make them unique?

```
uniq <- funspaceUniqueness(x=pca, axes=c(1,2))
traits_clean$uniqueness <- uniq
head(traits_clean[order(-uniq), c("Realm", "uniqueness")], 5)
```
